

User Manual

Interactive Data Reduction and EXAFS Analysis Dashboard

0. Prerequisites & Environment

To run these dashboards, ensure your environment meets the following requirements:

- **Software:** Python 3.11+ and JupyterLab or Jupyter Notebook.
- **Core Libraries:** xraylarch (for XAS processing), numpy, pandas, scipy, and h5py.
- **GUI Dependencies:** ipywidgets, matplotlib, and **ipyimpl**.
- **Plotting Backend:** For interactive widgets to function, you **must** use the ipyimpl backend by running:

```
from BS import *  
%matplotlib widget
```

- **Note for Older Environments:** If using JupyterLab 3.x or older, the following extensions may need to be installed manually via command line:

```
jupyter labextension install @jupyter-widgets/jupyterlab-manager jupyter-matplotlib
```

(Note: In modern 2024–2026 environments, these are typically pre-installed.)

1. BioXAS Data Reduction Dashboard

```
loader = XASLoader("project number") ;loader.gui()
```

BioXAS Data Reduction Dashboard

1. File Selection

Files

- Cr_E_Calibration_Scan_1.dat
- Cr_E_Calibration_Scan_2.dat
- Cr_Foil_E_Calibration_2.dat
- Cu_E_Calibration_Scan_1.dat
- Sample_Wheel_1_1.dat
- Sample_Wheel_1_Well_1_Nd_mica_4wk_1.dat
- Sample_Wheel_1_Well_1_Nd_mica_4wk_2.dat
- Sample_Wheel_1_Well_1_Nd_mica_4wk_3.dat
- Sample_Wheel_1_Well_1_Nd_mica_4wk_4.dat

2. Detector Settings

Element Line Board

3. Plotting & Processing

Mode ☒ Normalize ☒ Plot Sum Legend

4. Deglitch Options

Deglitch Offset

5. Export Options

Export Filename

Run Analysis & Plot

Figure 1: BioXAS Data Reduction Dashboard.

This GUI manages raw data ingestion and initial processing before EXAFS analysis.

Section 1: File Selection

- **Files List:** Click to highlight files. Use **Ctrl+Click** for multiple selections.
- **Interactive Inspection:** Clicking a filename triggers the 32-channel detector grid.
- **Channel Map:** Toggle channels by clicking their subplots. Green means the channel is active for summation; Red means it is excluded.

Section 2: Detector & Plotting Settings

- **Element/Line:** Defines the fluorescence window (e.g., Zn Kα1).

- **Board Dropdown:** Switches between **InB** and **OutB** detector banks.
- **Mode:** Choose **fluorescence** (summed channels / I_0), **transmission** ($-\ln(I_1/I_0)$), or **reference**.

Section 3: Deglitching & Execution

- **Deglitch Dropdown:** Select the region (**EXAFS**, **XANES**, or **Pre-edge**).
- **Offset:** Numerical value to vertically shift spectra in the multi-plot view.
- **Run Analysis:** Processes the selected files, applies the deglitcher, and generates the preview plot.

2. EXAFS Analysis Dashboard

```
toolkit = EXAFSToolkit('processed(x,mu)data.dat'); toolkit.gui()
```

Figure 2: EXAFS Analysis Dashboard.

This GUI handles the mathematical transformation of absorption data into K and R space.

Navigation Tabs

- **Mutran**: The primary view for $\mu(E)$. Use the **Mu View** dropdown to see **ALL** (with background/pre-edge overlays), **Normalized**, or **Flat**.
- **K-space**: Plots $\chi(k)$ weighted by the **k-weight** slider. The signal is automatically smoothed via a Savitzky-Golay filter for clarity.
- **R-space**: Displays the Fourier Transform magnitude. Control the x-axis range with the **R-max** slider.

- **Wavelet** : Performs a Cauchy Wavelet transform to visualize 2D Magnitude and Real Part plots.

Refinement Parameters

- **Pre-edge/Norm**: Adjust **Pre1/2** and **Norm1/2** to fix the baseline and edge-step normalization.
- **Background (rbkg)**: Sets the R -distance cutoff for the background spline.
- **FT Settings**: Define the transform window using **kmin**, **kmax**, and the **Window** type (Kaiser or Hanning).

Data Export

- **Export Type**: Select the data format (Normalized Mu, K-space, or R-space).
- **Filename**: Enter the output name and click **Export Data** to save a tab-separated file.